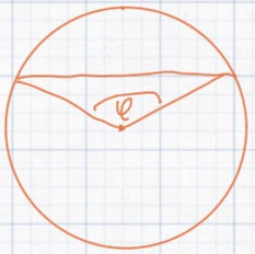
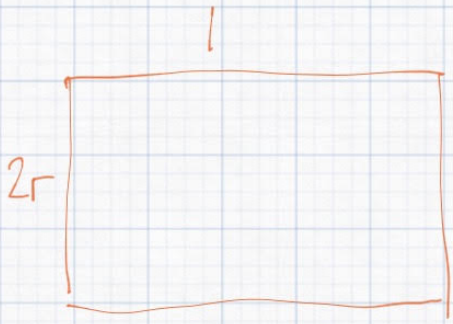
Aufgabe 3

a)





$$\frac{1}{2}r^2(\varphi - \sin\varphi) = \pi r^2 \cdot \frac{1}{4}$$

$$\frac{1}{2}r^2(\varphi - \sin\varphi) = \frac{1}{2}r^2 \cdot \pi\frac{2}{4}$$

$$\varphi - \sin\varphi = \pi\frac{1}{2}$$

$$\sin\varphi - \varphi = -\frac{1}{2}\pi$$

b) $f(x) = \sin x - x + \frac{1}{2}\pi = 0$

$F(x) = \sin x + \frac{1}{2}\pi = x$

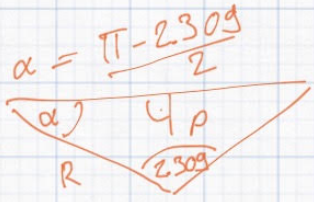
Startwert = 100° = 1.7 rad

n	x_n
0	1.7
1	2.5
2	2.1
...	
15	2.309

c) $h = r + p$

$h = r + 0.404r$

$h = 1.404r$



$\sin(\alpha) = \left(\frac{p}{r}\right)$

$$p = \sin\left(\frac{\pi - 2.309}{2}\right) \cdot r$$

$$p = 0.404r$$